```c
#include <stdio.h>

int main()
{
    int a,b;
    printf("enter a numbers: ");
    scanf("%d%d",&a,&b);

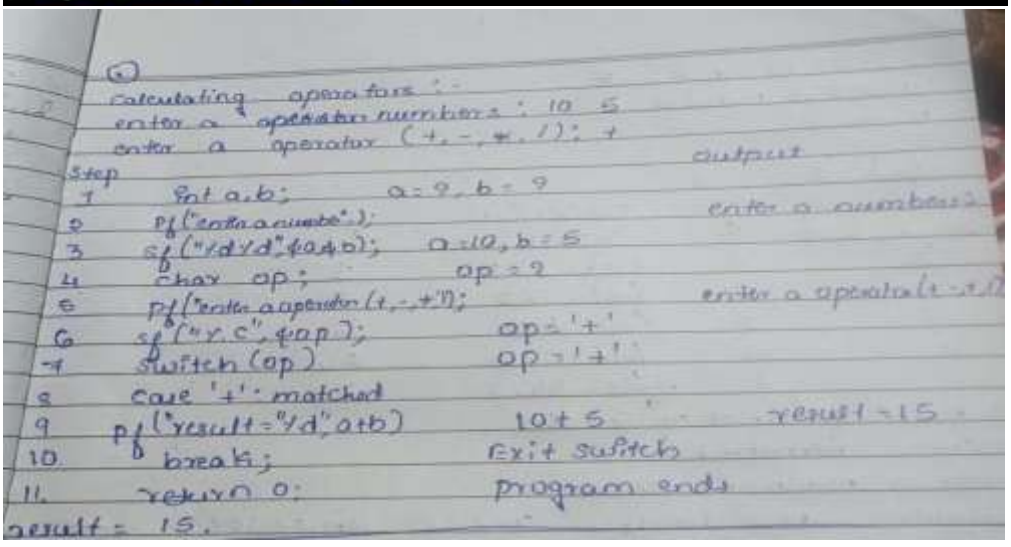
    char op;
    printf("enter a operator(+,-,*,/): ");
    scanf(" %c", &op);

    switch(op)
    {
        case '+':
            printf(" result= %d",a+b);
            break;
        case '-':
            printf("result= %d",a-b);
            break;
        case '*':
            printf(" result= %d",a*b);
            break;
        case '/':
           if(b!=0)
            printf(" result= %d",a/b);
            else
            printf("not divisible by zero");
            break;
        default:
            printf("Invalid operator");
           }

    return 0;
}
```

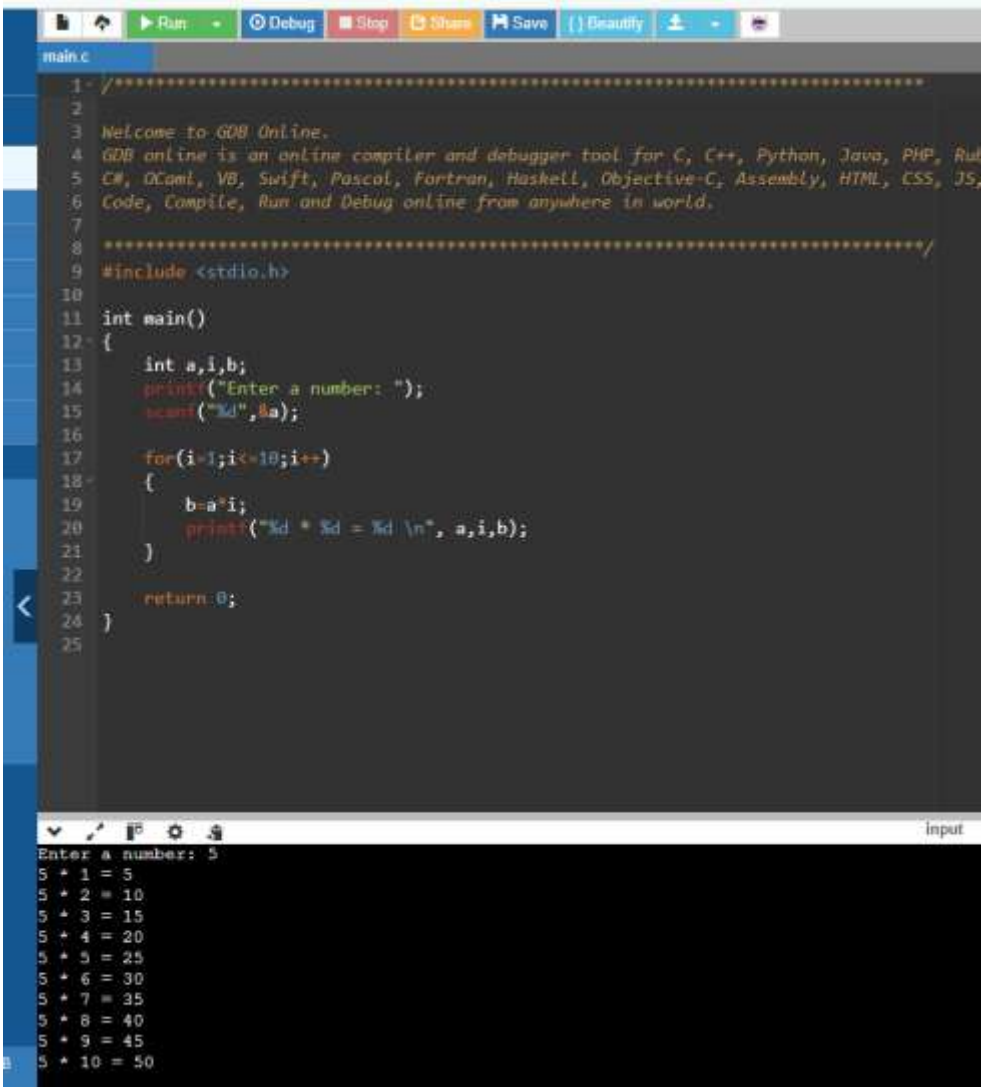
```
Input
enter a numbers: 6 8
enter a operator(+,-,*,/): +
 result= 14

...Program finished with exit code 0
```



Calculating operations

multiplication table program



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

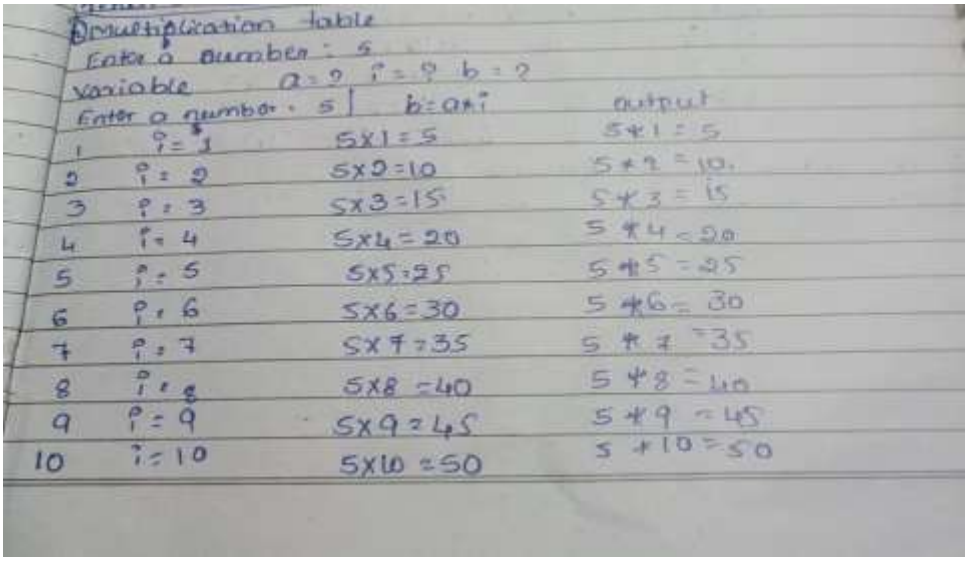
int main()
{
    int a,i,b;
    printf("Enter a number: ");
    scanf("%d",&a);

    for(i=1;i<=10;i++)
    {
        b=a*i;
        printf("%d * %d = %d \n", a,i,b);
    }

    return 0;
}
```

Input

```
Enter a number: 5
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
```



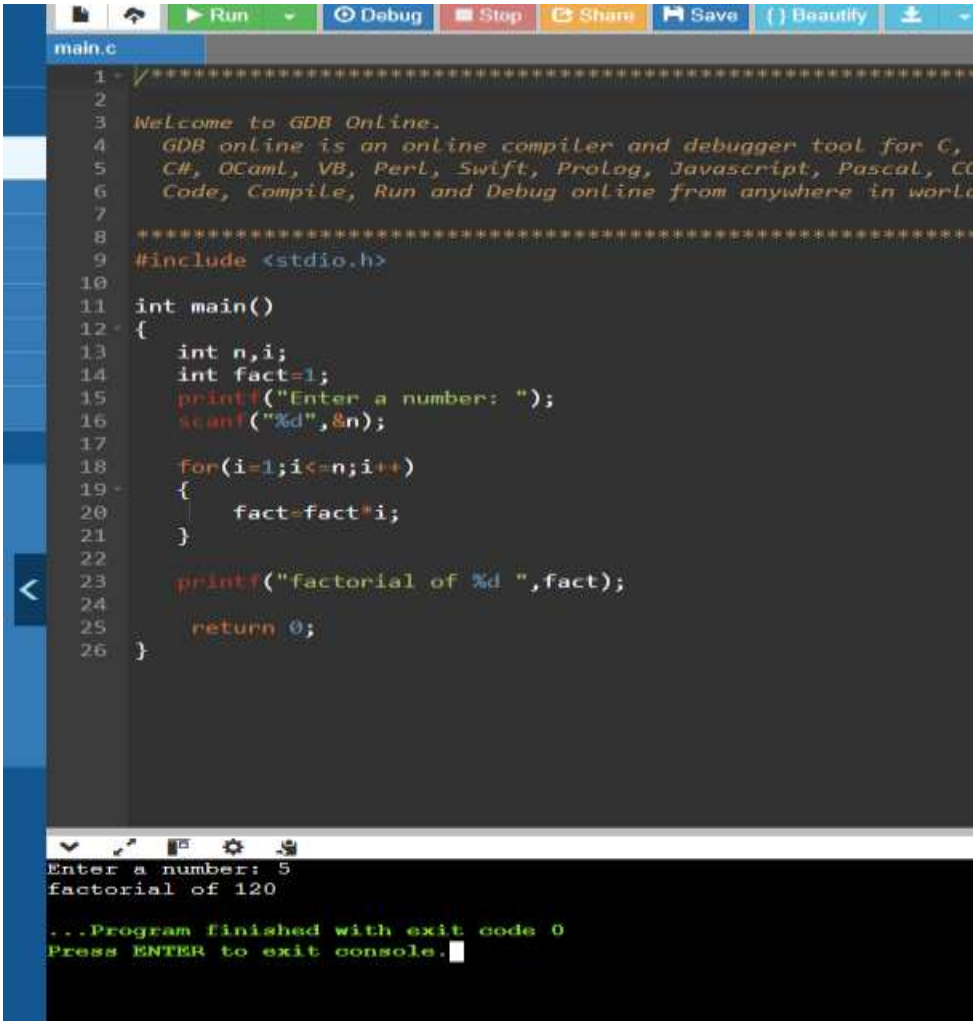
Multiplication table

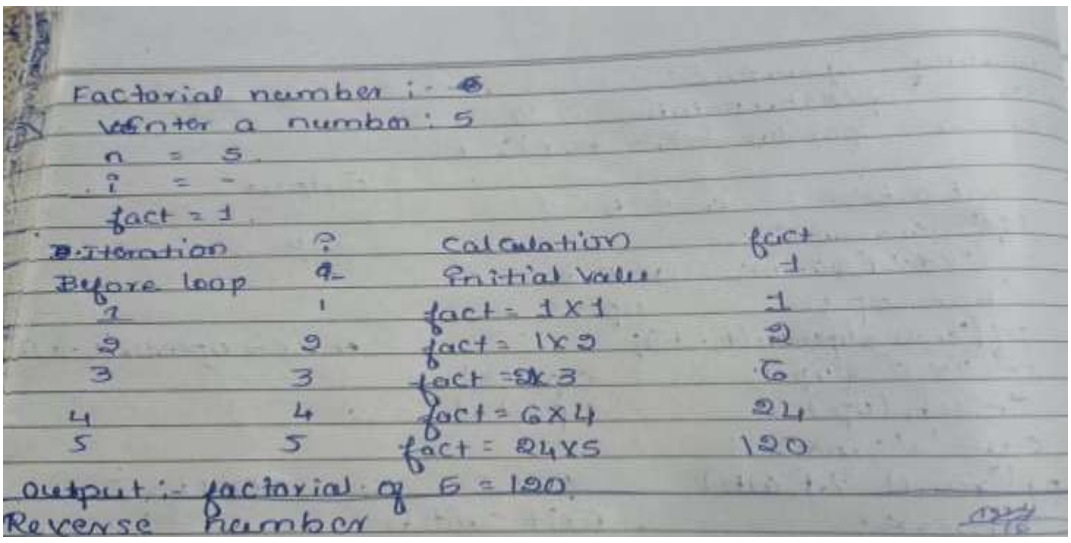
Enter a number: 5

Variable a=? i=? b=?

	Enter a number: 5	b=a*i	output
1	i=1	5x1=5	5*1=5
2	i=2	5x2=10	5*2=10
3	i=3	5x3=15	5*3=15
4	i=4	5x4=20	5*4=20
5	i=5	5x5=25	5*5=25
6	i=6	5x6=30	5*6=30
7	i=7	5x7=35	5*7=35
8	i=8	5x8=40	5*8=40
9	i=9	5x9=45	5*9=45
10	i=10	5x10=50	5*10=50

factorial program





Reverse number

main.c

```c
/******************************************************

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******************************************************/
#include <stdio.h>

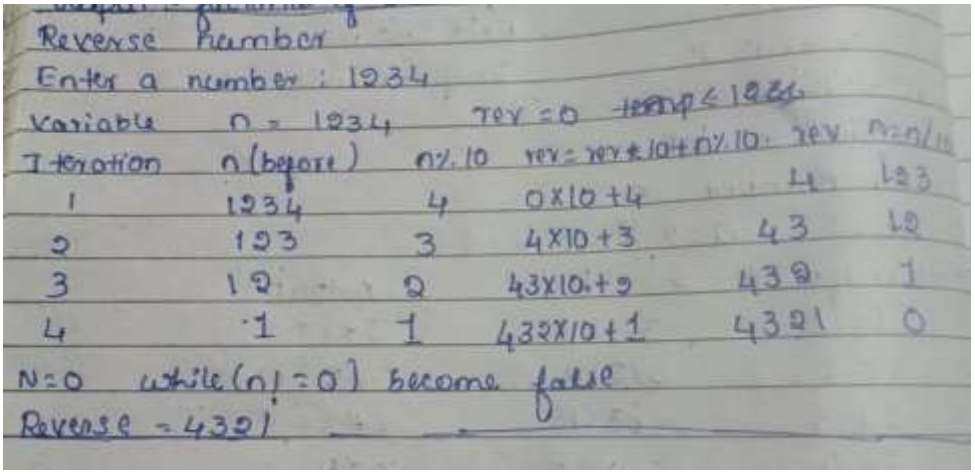
int main()
{
   int n ,rev=0;
   printf("Enter a number: ");
   scanf("%d",&n);

   while(n!=0)
   {
       rev=rev*10+ n%10;
       n=n/10;
   }
   printf("Reverse = %d\n",rev);

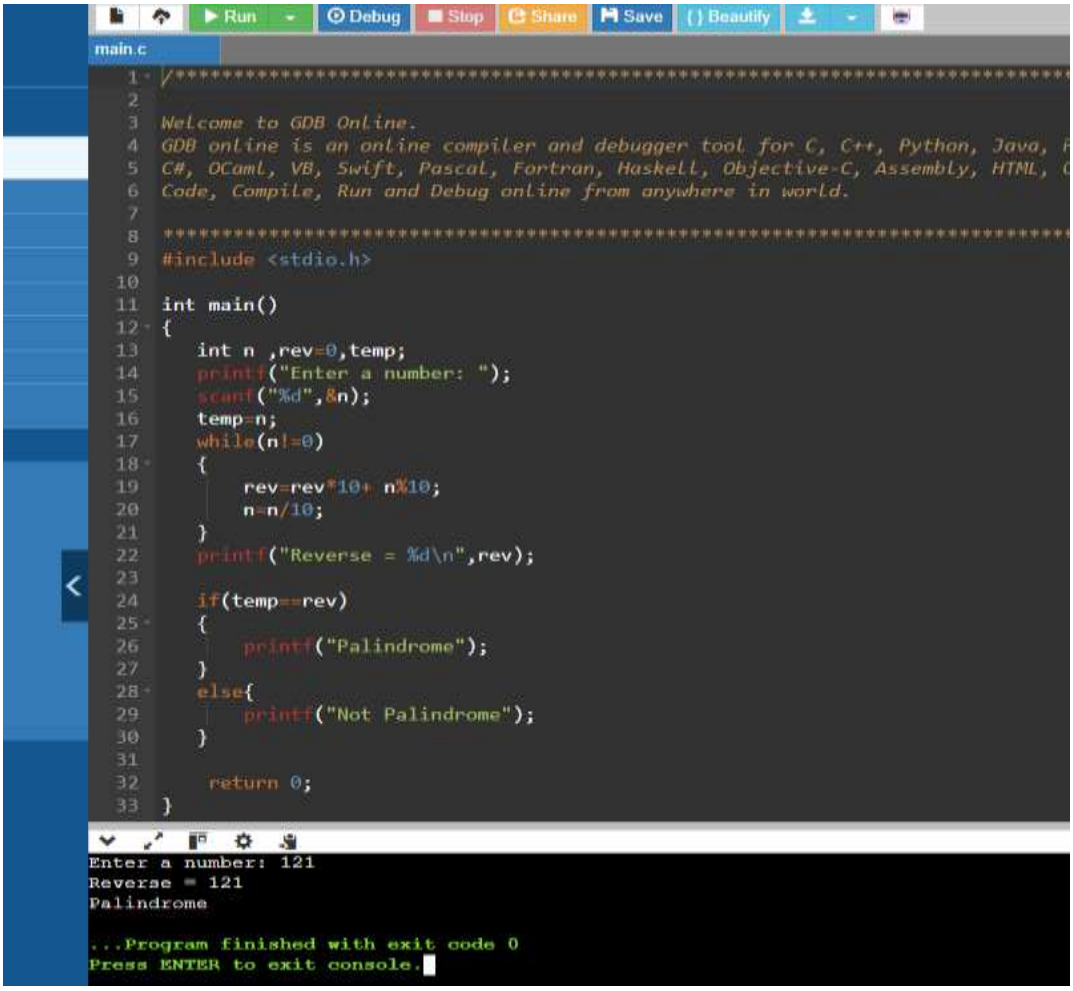

    return 0;
}
```

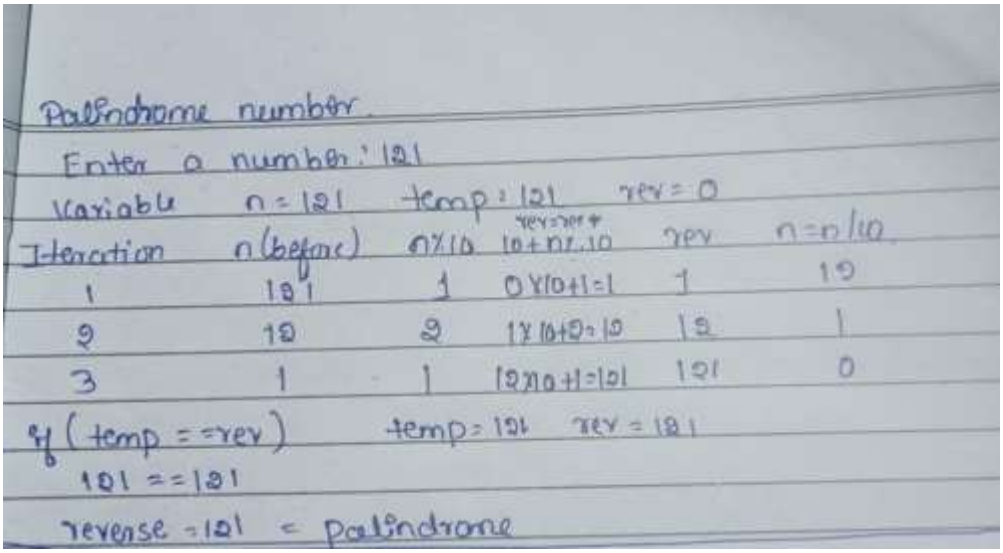
```
Enter a number: 567
Reverse = 765


...Program finished with exit code 0
Press ENTER to exit console.
```

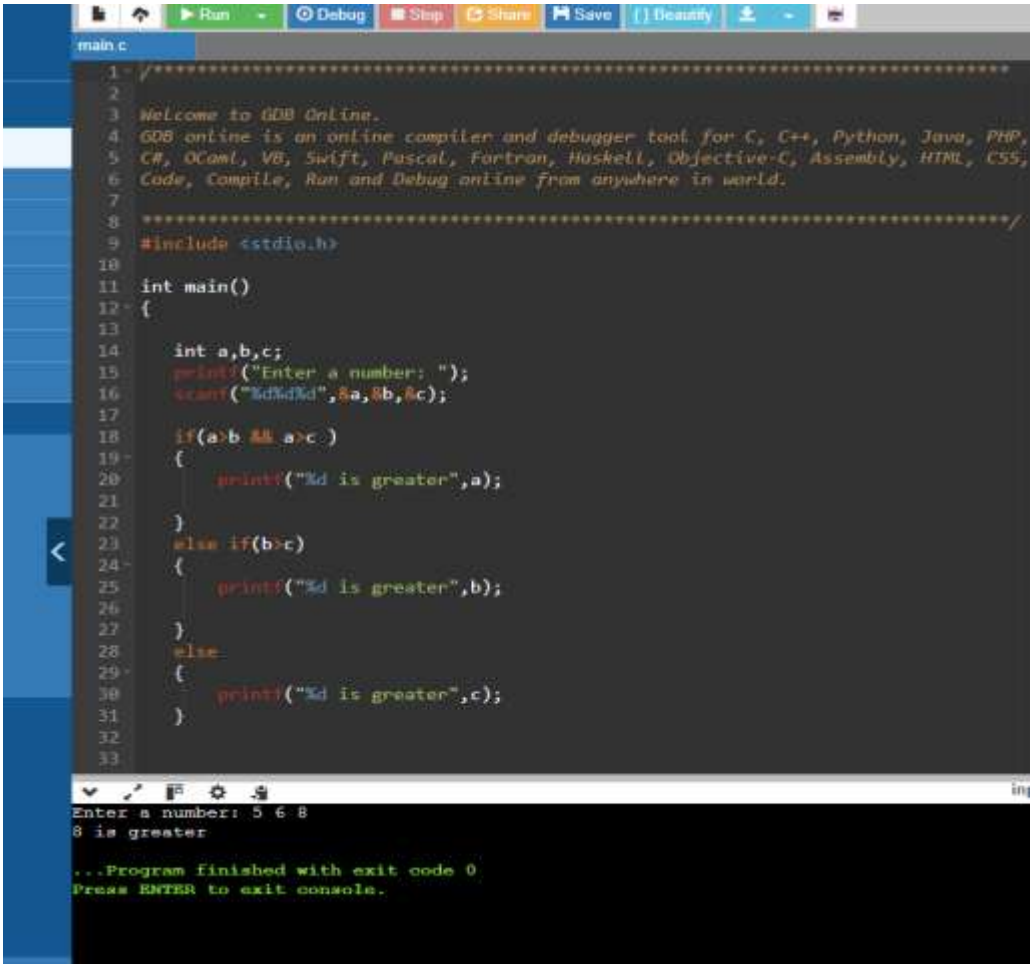


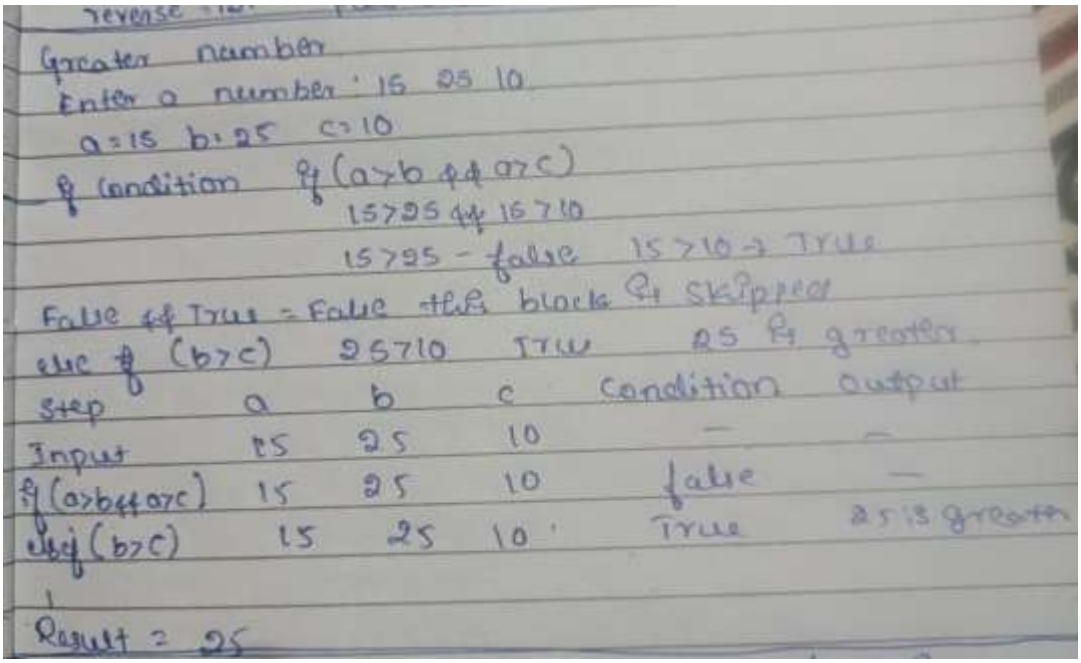
Palindrome number



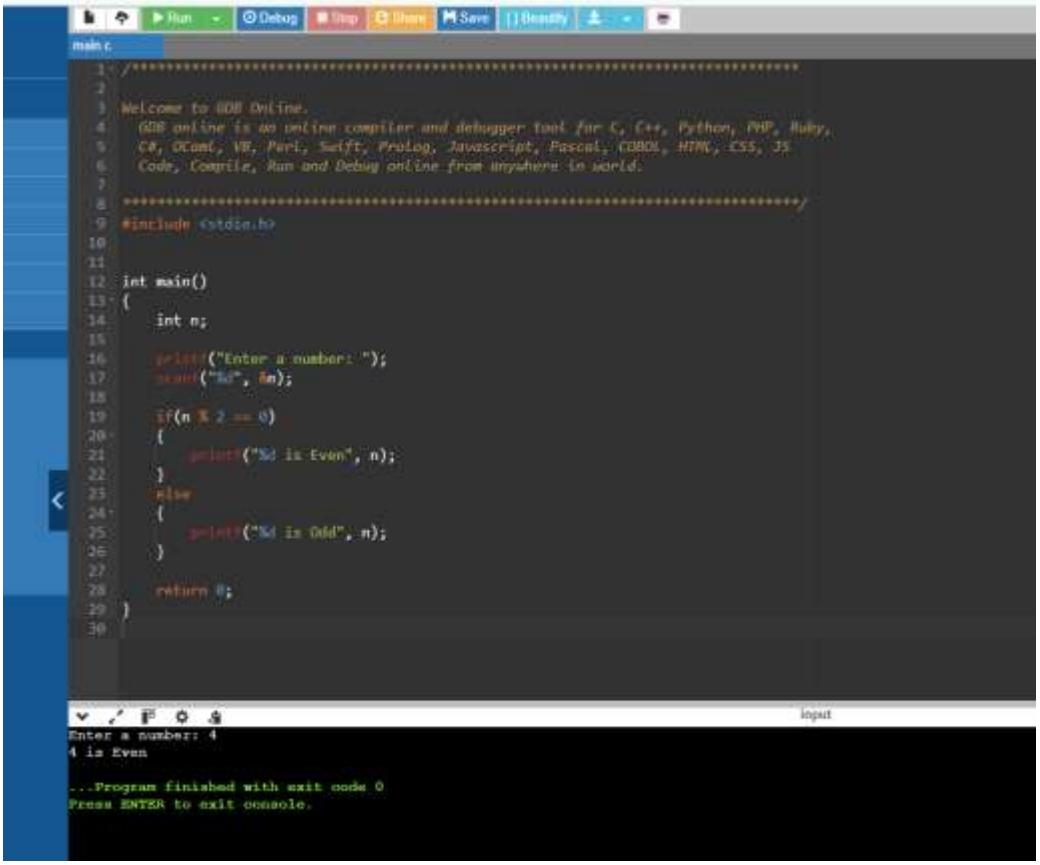


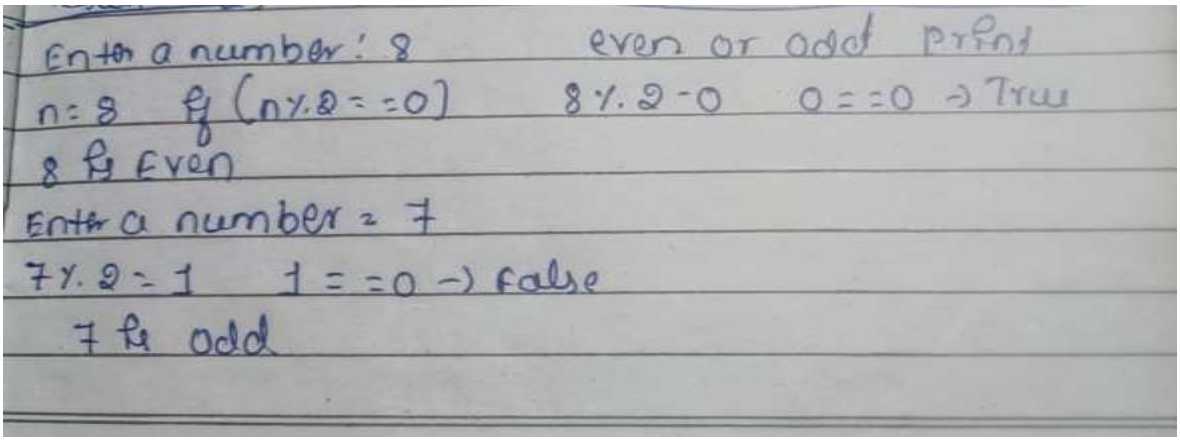
Greater number





Even or odd number





Adding two numbers using functions

Run Debug Stop Share Save {} Beautify

main.c

```c
/*****************************************************************************

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C#, OCaml, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly,
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*****************************************************************************/
#include <stdio.h>

int add(int a, int b)
{
    return a + b;
}

int main()
{
    int x, y, sum;

    printf("Enter two numbers: ");
    scanf("%d%d", &x, &y);

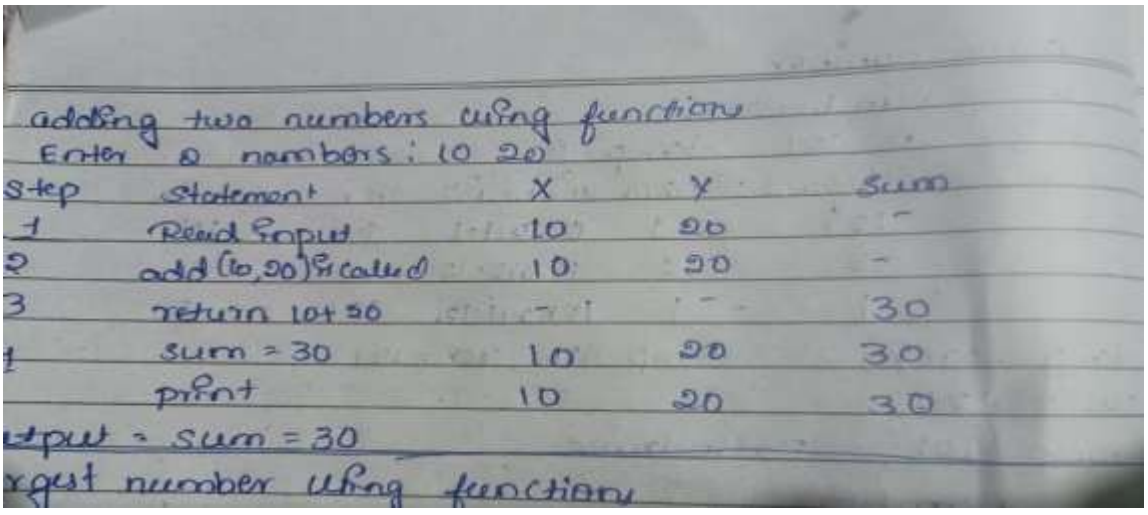
    sum = add(x, y);

    printf("Sum = %d", sum);

    return 0;
}
```

```
Enter two numbers: 5 6
Sum = 11

...Program finished with exit code 0
Press ENTER to exit console.
```



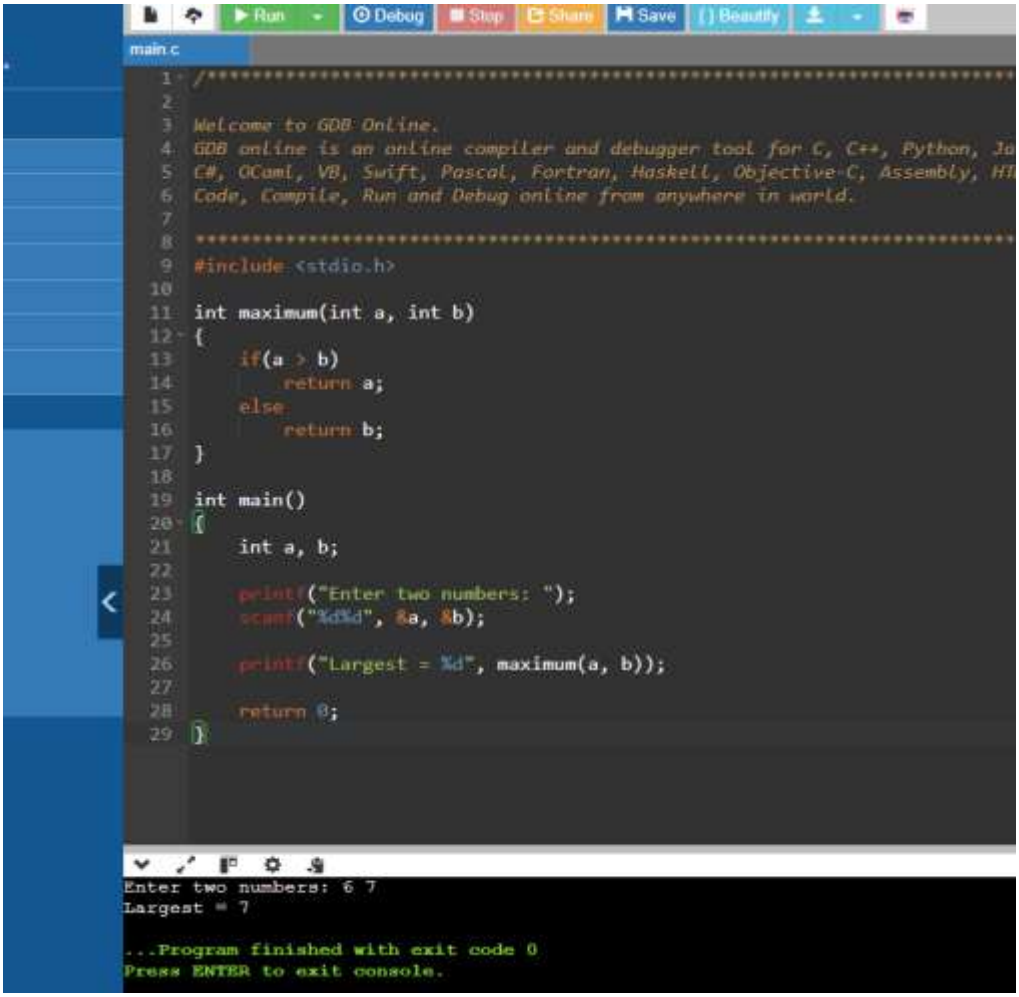
adding two numbers using functions
Enter 2 numbers: 10 20

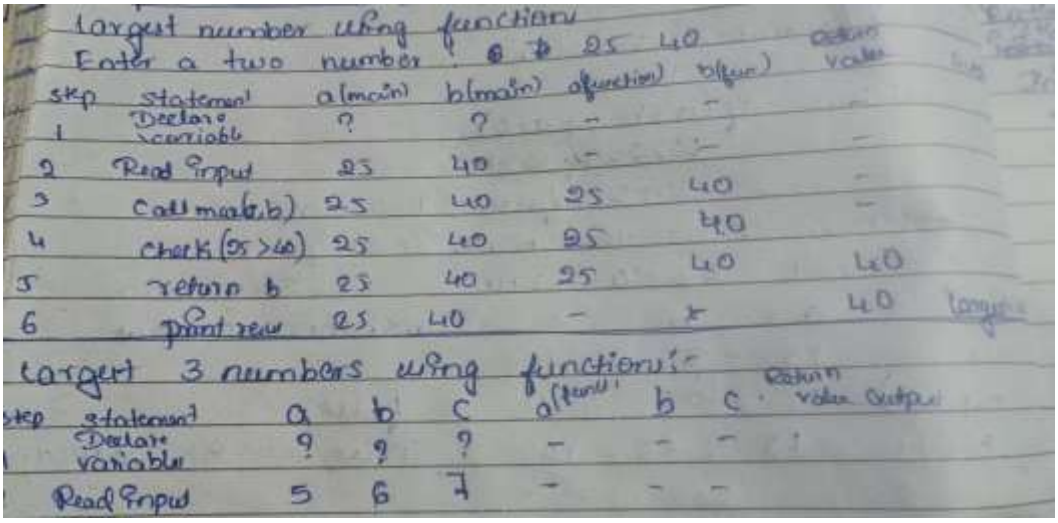
Step	Statement	X	Y	Sum
1	Read Input	10	20	-
2	add(10,20) is called	10	20	-
3	return 10+20	-	-	30
4	sum = 30	10	20	30
	print	10	20	30

utput : sum = 30

rgest number using functions

Largest number using functions





Largest two number using functions

```c
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int max3(int a, int b, int c)
{
    if(a >= b && a >= c)
        return a;
    else if(b >= c)
        return b;
    else
        return c;
}

int main()
{
    int a, b, c;

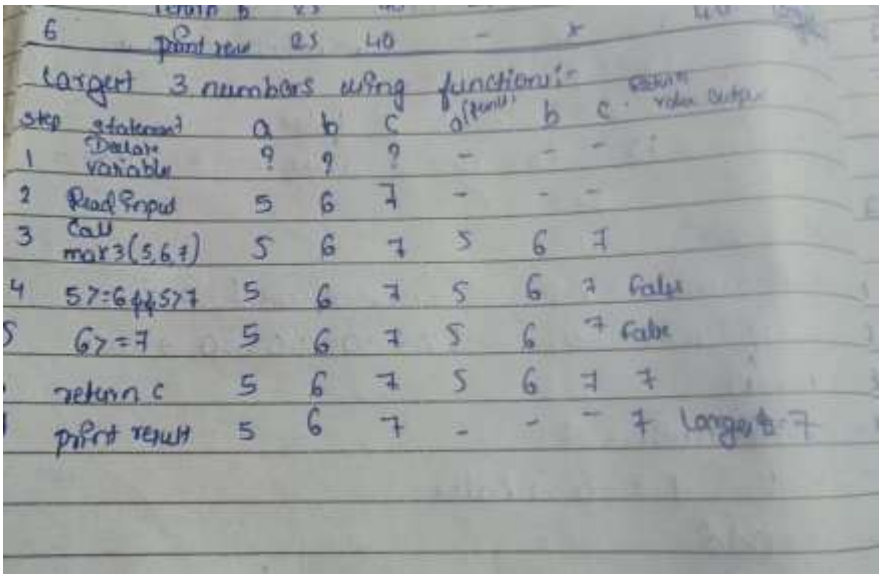
    printf("Enter three numbers: ");
    scanf("%d%d%d", &a, &b, &c);

    printf("Largest = %d", max3(a, b, c));

    return 0;
}
```

```
Enter three numbers: 5 6 7
Largest = 7

...Program finished with exit code 0
Press ENTER to exit console.
```



Check wether Positive or negative numbers

```c
/*****************************************************************************

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C#, OCaml, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HT
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*****************************************************************************/
#include <stdio.h>

void check(int n)
{
    if(n > 0)
        printf("Positive");
    else if(n < 0)
        printf("Negative");
    else
        printf("Zero");
}

int main()
{
    int n;

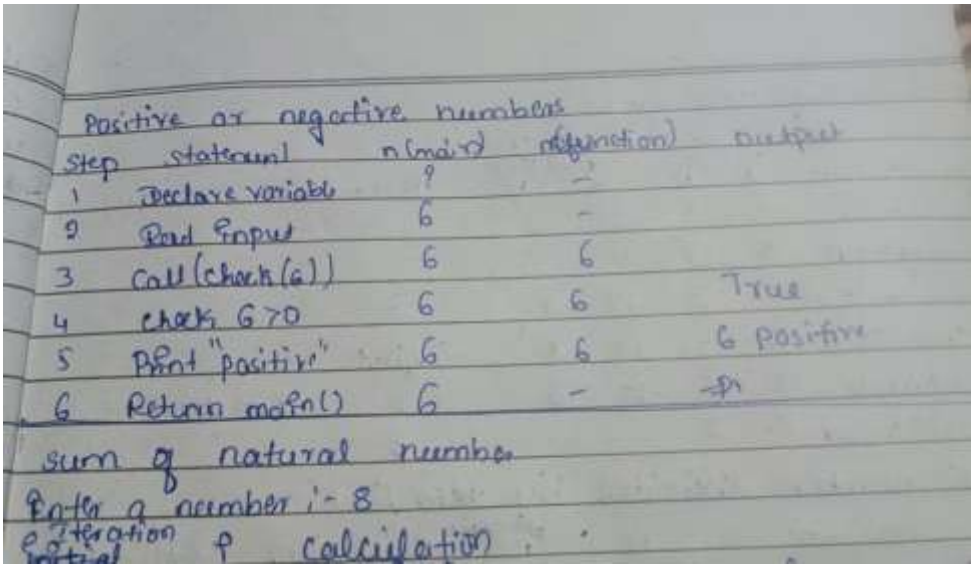
    printf("Enter a number: ");
    scanf("%d", &n);

    check(n);

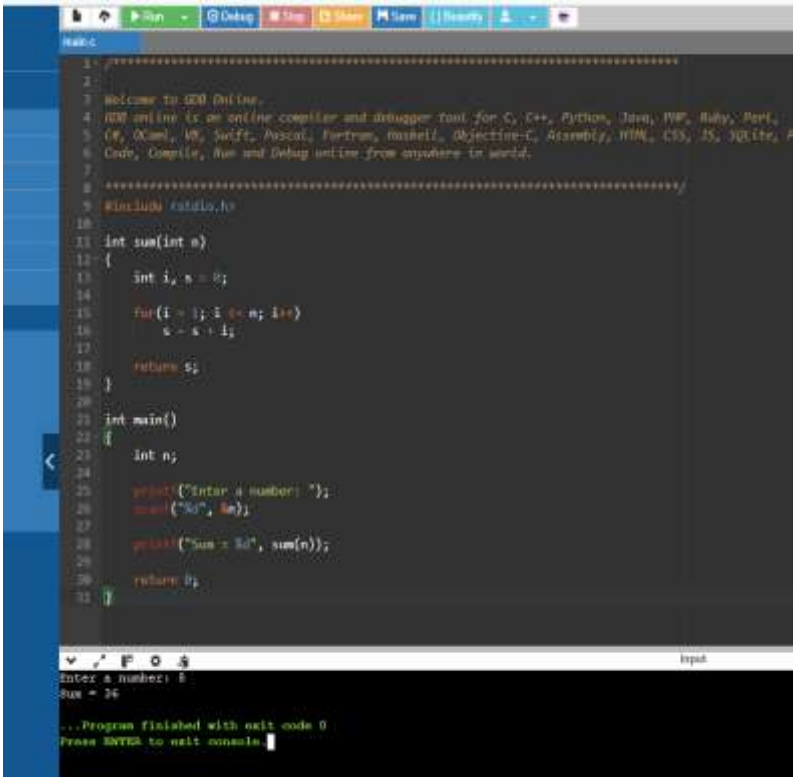
    return 0;
}
```

```
Enter a number: 6
Positive

...Program finished with exit code 0
Press ENTER to exit console.
```



Sum of natural numbers



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int sum(int n)
{
    int i, s = 0;

    for(i = 1; i <= n; i++)
        s = s + i;

    return s;
}

int main()
{
    int n;

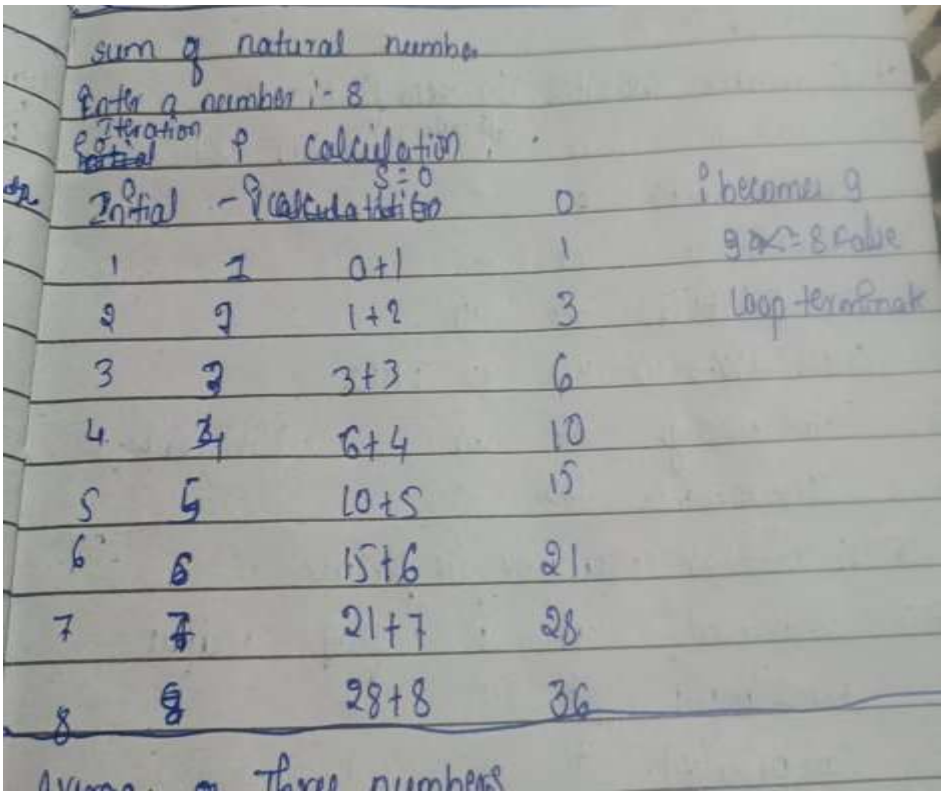
    printf("Enter a number: ");
    scanf("%d", &n);

    printf("Sum = %d", sum(n));

    return 0;
}
```

```
Enter a number: 8
Sum = 36

...Program finished with exit code 0
Press ENTER to exit console.
```



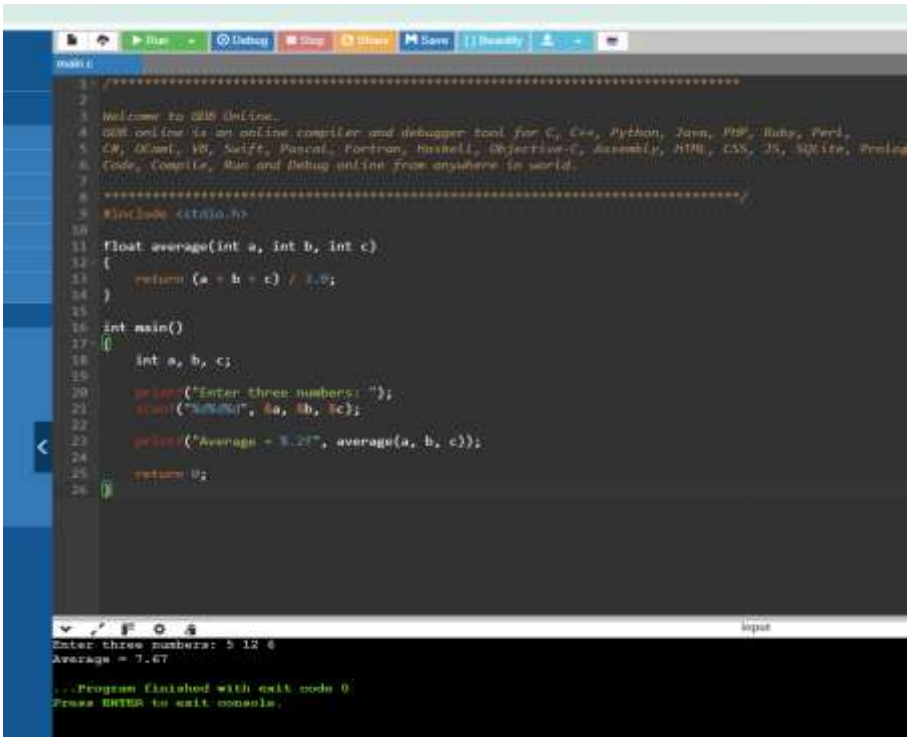
sum of natural numbers

Enter a number :- 8

Initial	Iteration & calculation	s := 0		
Initial	-		0	i becomes 9
1	1	0+1	1	9 <= 8 false
2	2	1+2	3	loop terminate
3	3	3+3	6	
4	4	6+4	10	
5	5	10+5	15	
6	6	15+6	21	
7	7	21+7	28	
8	8	28+8	36	

Three numbers

Average of three numbers



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

float average(int a, int b, int c)
{
    return (a + b + c) / 3.0;
}

int main()
{
    int a, b, c;

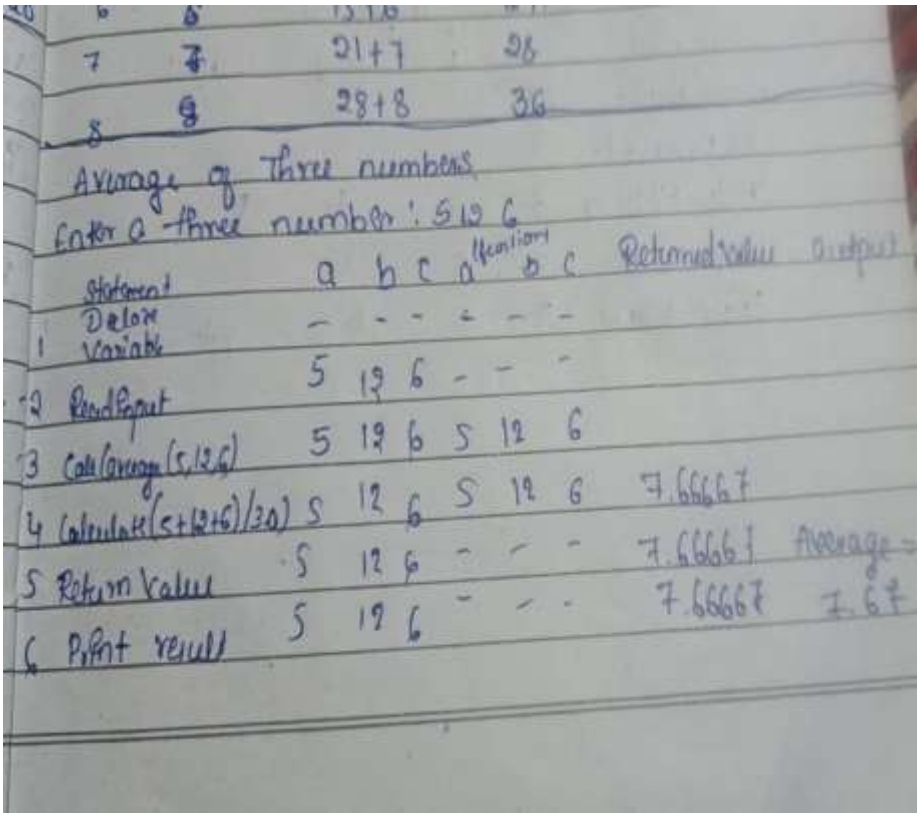
    printf("Enter three numbers: ");
    scanf("%d%d%d", &a, &b, &c);

    printf("Average = %.2f", average(a, b, c));

    return 0;
}
```

```
Enter three numbers: 5 12 6
Average = 7.67

...Program finished with exit code 0
Press ENTER to exit console.
```

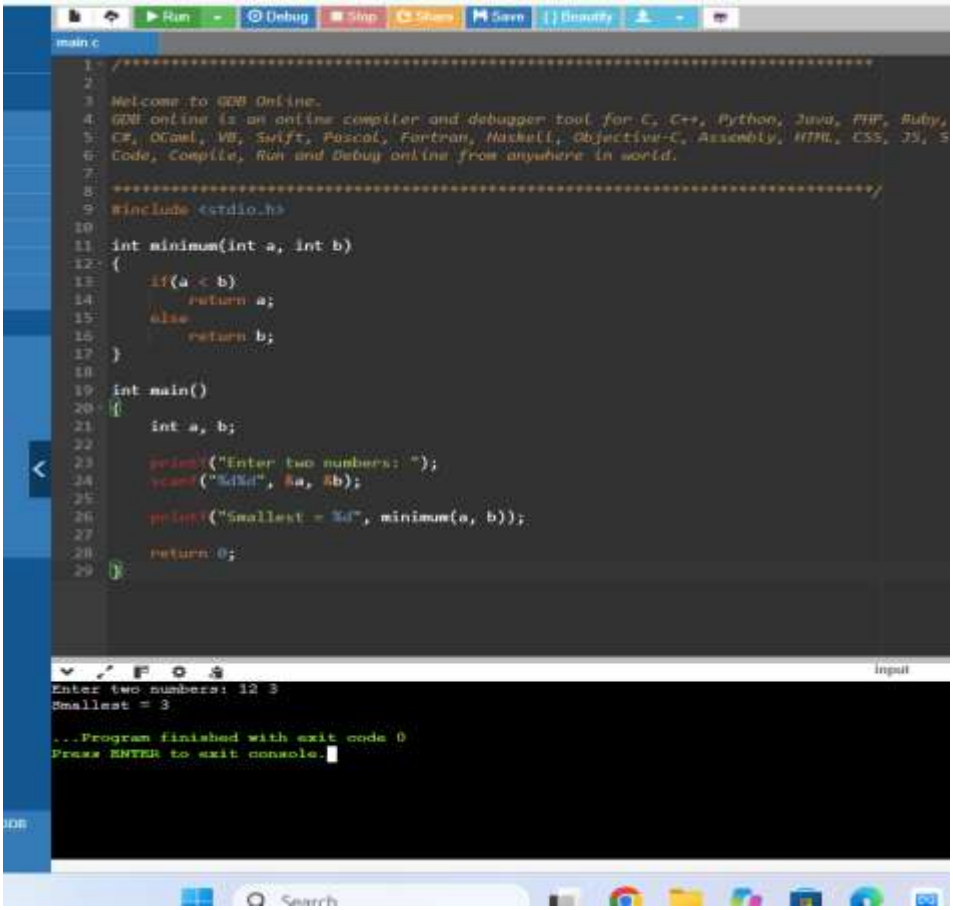


Average of Three numbers

Enter a three number: 5 12 6

	Statement / Data / Variable	a	b	c	a (function)	b	c	Returned value	Output
1		-	-	-	-	-	-		
2	Read Input	5	12	6	-	-	-		
3	Call (average (5,12,6)	5	12	6	5	12	6		
4	(a+b+c)/3.0	5	12	6	5	12	6	7.66667	
5	Return value	5	12	6	-	-	-	7.66667	Average =
6	Print result	5	12	6	-	-	-	7.66667	7.67

Minimum of two numbers



```
/*****************************************************************************

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C#, OCaml, VB, Swift, Pascal, Fortran, Haskell, Objective-C, Assembly, HTML, CSS, JS, S
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*****************************************************************************/
#include <stdio.h>

int minimum(int a, int b)
{
    if(a < b)
        return a;
    else
        return b;
}

int main()
{
    int a, b;

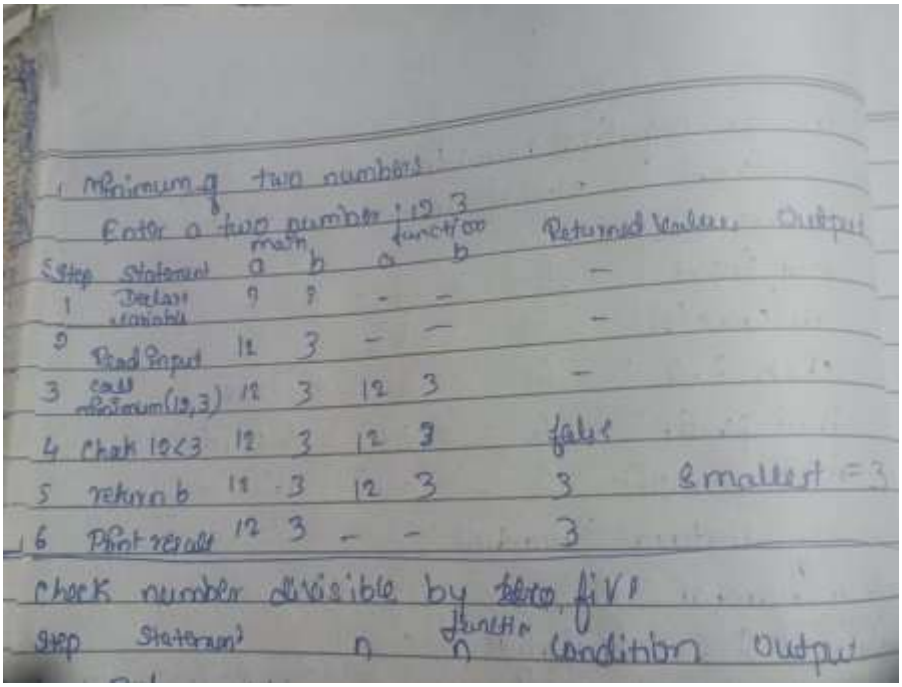
    printf("Enter two numbers: ");
    scanf("%d%d", &a, &b);

    printf("Smallest = %d", minimum(a, b));

    return 0;
}
```

```
Input
Enter two numbers: 12 3
Smallest = 3

...Program finished with exit code 0
Press ENTER to exit console.
```



1. Minimum of two numbers

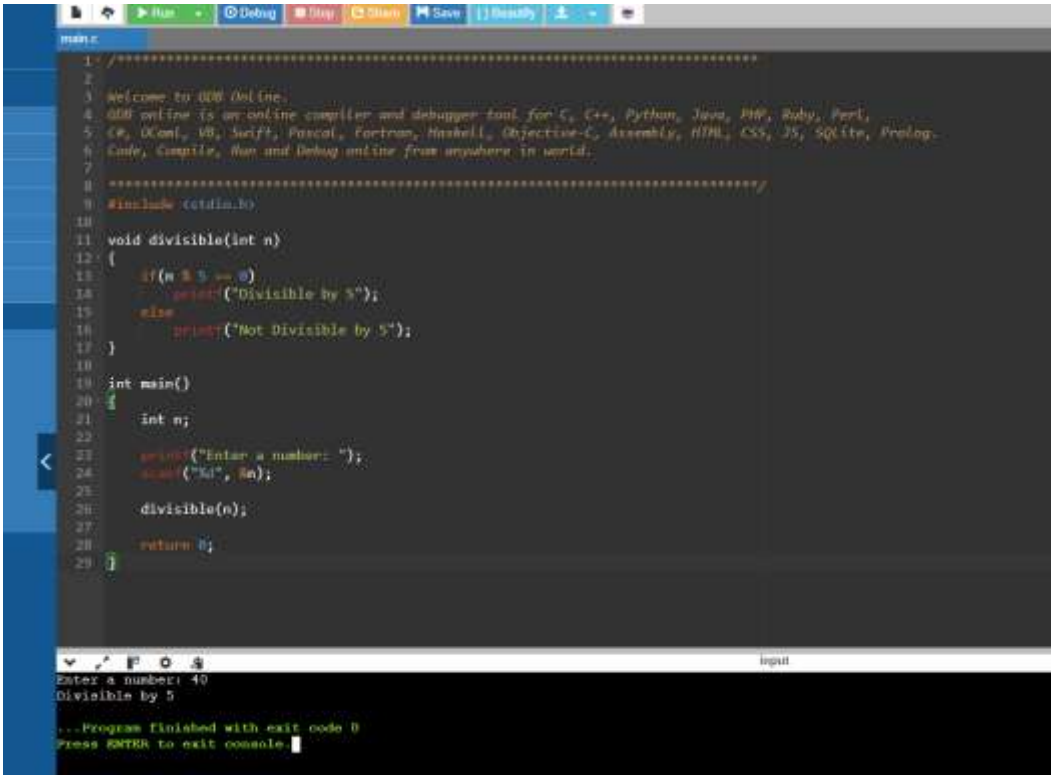
Enter a two number : 12 3

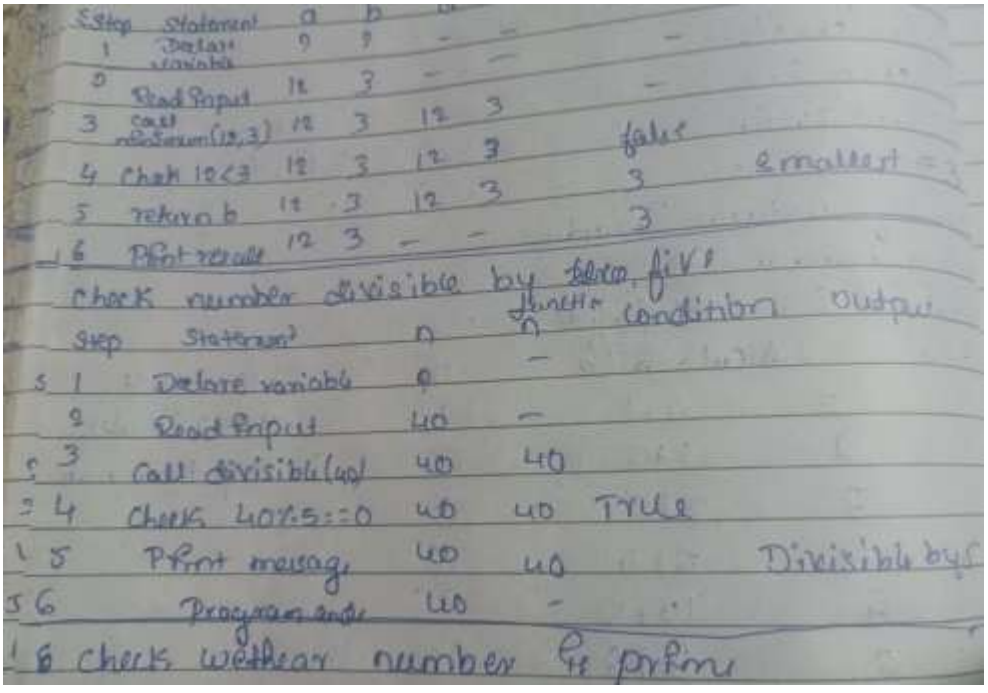
Step	Statement	main a	main b	function a	function b	Returned values	Output
1	Declare variable	?	?	-	-	-	
2	Read Input	12	3	-	-	-	
3	call minimum(12,3)	12	3	12	3	-	
4	Check 12<3	12	3	12	3	false	
5	return b	12	3	12	3	3	&mallest = 3
6	Print result	12	3	-	-	3	

check number divisible by two, five

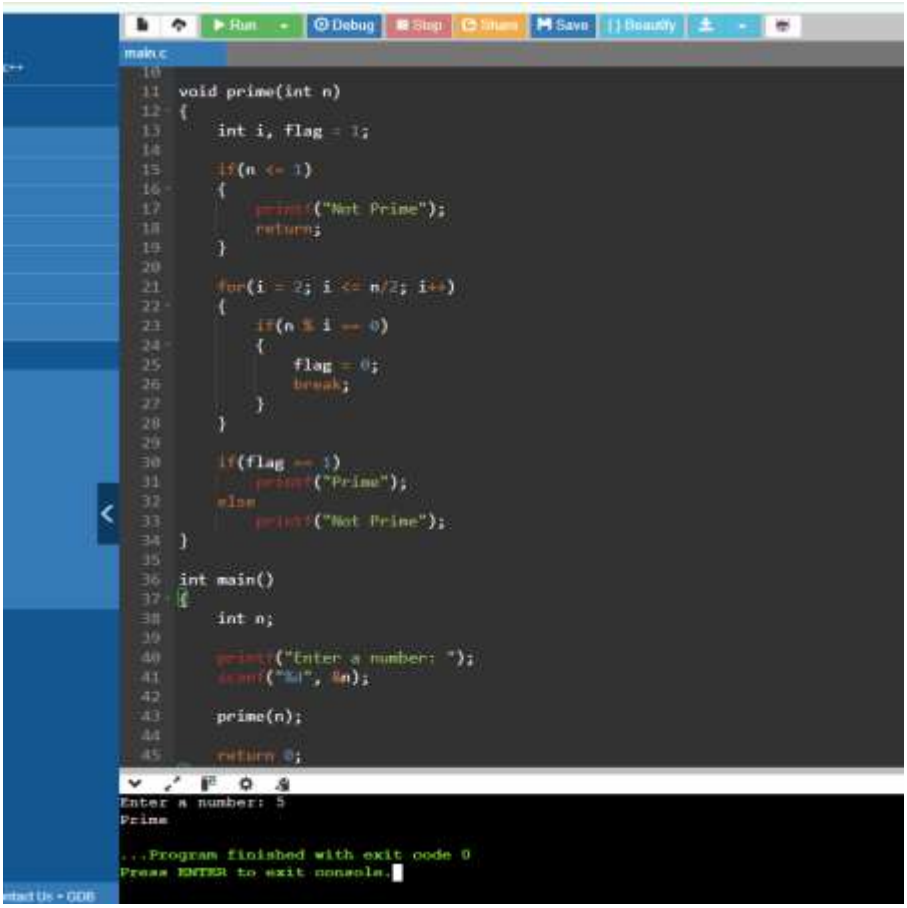
Step	Statement	n	function n	condition	Output

Check number is divisible by 5





Check wether number is prime



```
Run  Debug  Stop  Share  Save  {} Beautify
main.c

void prime(int n)
{
    int i, flag = 1;

    if(n <= 1)
    {
        printf("Not Prime");
        return;
    }

    for(i = 2; i <= n/2; i++)
    {
        if(n % i == 0)
        {
            flag = 0;
            break;
        }
    }

    if(flag == 1)
        printf("Prime");
    else
        printf("Not Prime");
}

int main()
{
    int n;

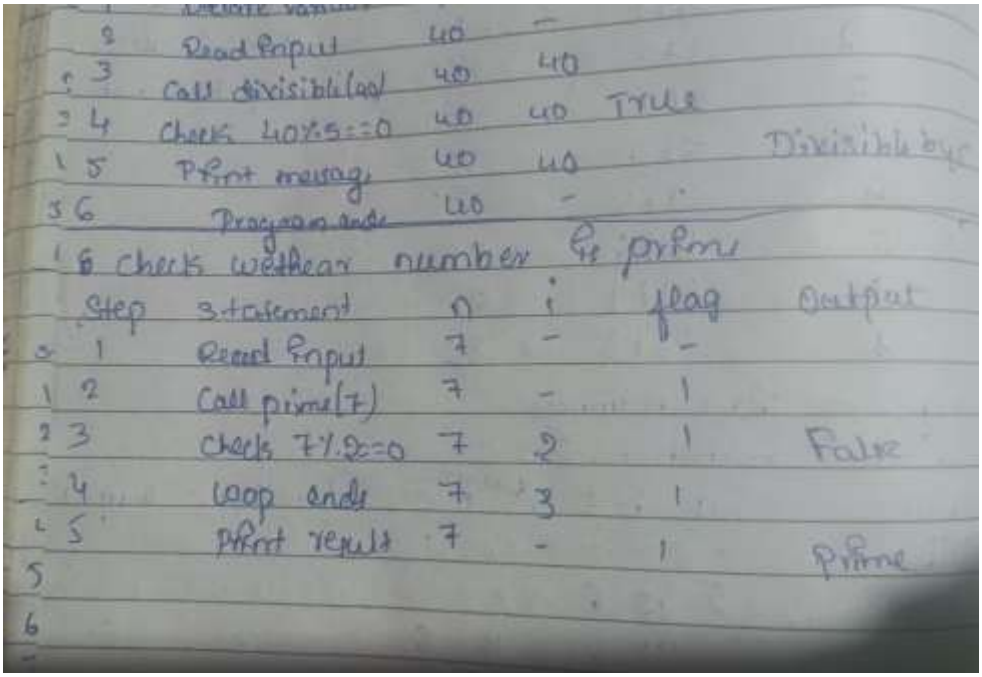
    printf("Enter a number: ");
    scanf("%d", &n);

    prime(n);

    return 0;
}

Enter a number: 5
Prime

...Program finished with exit code 0
Press ENTER to exit console.
```



	Step	statement	n	i	flag	output
1	1	Read input	40	-	-	
2	2	Call divisible(40)	40	40		
3	3	Check 40%5==0	40	40	TRUE	
4	4	Print message	40	40		Divisible by 5
5	5	Program ends	40	-		

Check wether number is prime

	Step	statement	n	i	flag	output
1	1	Read input	7	-	-	
2	2	Call prime(7)	7	-	1	
3	3	check 7%2==0	7	2	1	False
4	4	loop ends	7	3	1	
5	5	Print result	7	-	1	Prime
6						